

AWS EC2 – Create Instance and Assign Elastic IP

Task:

Create an EC2 instance and assign an Elastic IP so that the public IP does not change after restart.

(a) Create EC2 Instance:

1. Login to AWS Management Console
 - Open <https://aws.amazon.com/>
 - Sign in to your account
2. Open EC2 Dashboard
 - Search and select **EC2**
3. Click **Launch Instance**
4. Enter Instance Name
 - Example: StaticIPServer
5. Choose AMI
 - Select **Ubuntu Server 22.04 LTS** or **Amazon Linux 2**
6. Choose Instance Type
 - Select **t2.micro** (Free Tier)
7. Create Key Pair
 - Name: **mykey**
 - Format: **.pem**
 - Download and save securely
8. Configure Security Group
 - Allow **SSH (Port 22)**
 - Allow **HTTP (Port 80)** (optional)
9. Launch Instance
 - Wait until instance is in **Running state**

(b) Allocate Elastic IP:

1. Go to EC2 Dashboard
2. In left menu, click **Elastic IPs**
3. Click **Allocate Elastic IP Address**
4. Click **Allocate**

(c) Associate Elastic IP with EC2 Instance:

1. Select the allocated Elastic IP
2. Click **Actions** → **Associate Elastic IP**
3. Choose:
 - Instance: Select your EC2 instance (`StaticIPServer`)
 - Private IP: Default selected
4. Click **Associate**

(d) Verify Elastic IP:

1. Go to **EC2 Instances**
2. Check Public IPv4 address
3. It will now show the **Elastic IP**

(e) Test IP Stability:

1. Restart EC2 instance
2. Check public IP again
3. IP remains same (Elastic IP does not change)

Example:

- Instance Name: StaticIPServer
- Elastic IP: 3.110.25.60
- OS: Ubuntu / Amazon Linux
- Type: t2.micro

Result:

- EC2 instance created successfully
- Elastic IP assigned and associated
- Public IP remains fixed even after restart